

Lesson 18 — Variables Driving Decisions & State

Matches LEGO Teacher Portal lesson: "Nap Needs".

Objective (I can...): I can use a variable inside a condition to make my program change its behavior based on state. **CS concept:** Variables combined with conditionals; the idea of program *state* · **Builds on:** Lesson 17 (updating) and Lesson 12 (if/then/else).

AI Connection (2 min): A model acts on stored data: IF accuracy on test examples $\geq$ target, THEN "trained enough." *First-principles question:* "What number would honestly tell you a system is 'good enough' to use?"

Materials: LEGO CS & AI kit + device per pair, trigger object/button, notebooks. **Model reuse:** Variables model from Lesson 16 — code only.

60-minute flow - Warm-up (8): "State changes everything" — coach describes a machine behaving differently at count 0 vs. count 5; pairs name the "state" that caused the change. Introduce *state*. -

Concept + demo (10): Demo: count events; if the score reaches a target, then switch behavior (celebration mode); else keep counting. Point out the variable is the program's memory of its state. -

Build/Code (30): Pairs code a counter plus an if/then that changes behavior when the variable hits a chosen value, and the behavior should visibly persist (stay in the new state). They note in the notebook what "state" their program is in before and after. *Swap at 15 min.* - **Challenge (7):** Add an else branch for "not yet," and a second threshold for a third state. - **Share + pack-up (5):** Pairs demo crossing the threshold and the state change, and explain what the variable "remembers." Model stays built.

Pair roles: Driver codes the if/then, Navigator owns the counter and narrates the state. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — provide the counter; students add the condition from framed pseudocode. *Extension / fast-finisher* — three states from two thresholds. *Reteach* — set the target to 2 so it triggers fast, then raise it.

Exit check: Pair's model changes and holds a new behavior when the variable reaches the target, and they can describe the state. **At-home (optional):** Describe a device with clear "states" (phone: locked / unlocked / low-battery); name what variable tracks each. ~10 min.